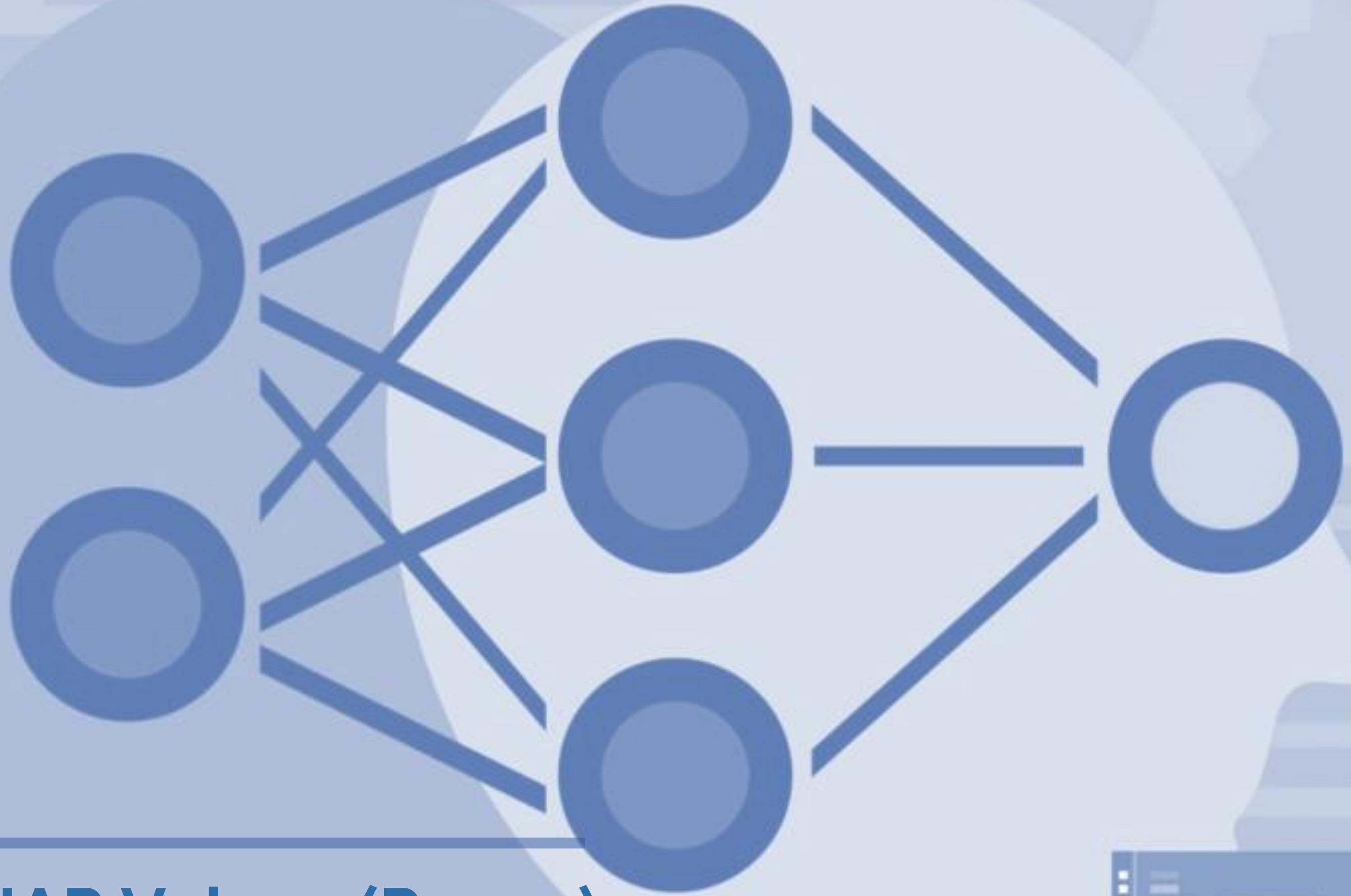


STAT606

Interpreting SHAP Values (Prac 5)

SHAP Summary (Beeswarm) Plot

- This plot gives a *global view* of how each variable influences predictions across the whole dataset (not just one observation).
- SHAP values are a way to explain *individual* predictions from a machine learning model by fairly attributing how much each feature contributed to that prediction.
- In other words, SHAP measures how much did each feature push the prediction away from the average prediction. So, each feature gets a positive or negative contribution.
- It is calculated by
  1. Taking every possible subset of features
  2. Adding feature i to that subset
  3. Measuring how much the prediction changes
  4. Averaging all those changes with weights
- Exact SHAP is computationally infeasible for large models, so random samples of subsets are taken instead.

How SHAP is actually computed in practice:

Different approximations are used based on the model:

a) Tree SHAP (for Random Forest, XGBoost, GBM etc.)

- Uses tree structure to compute exact Shapley values efficiently
- Very fast and exact for tree models

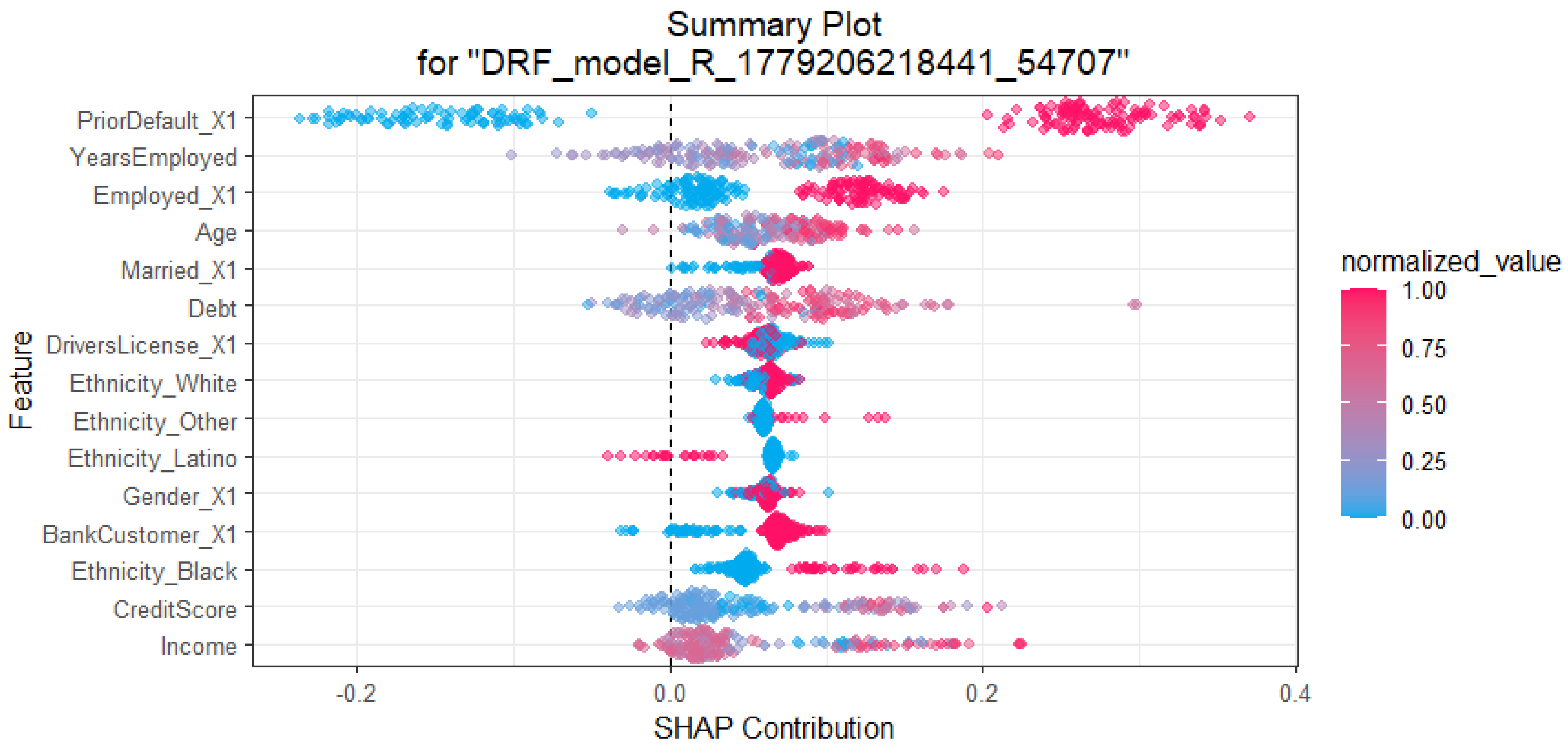
b) Kernel SHAP (model-agnostic)

- Uses regression on sampled coalitions
- Approximates Shapley values
- Slower but works for any model

c) Deep SHAP

- Designed for neural networks
- Combines SHAP with backprop-style approximations

Let's consider an example generated from prac 5:



What a SHAP Beeswarm Plots Reveal (Summary)

Insight	How to Read It from the Plot
Feature Importance Ranking:	Vertical order (top = most important, bottom = least important)
Magnitude of Impact:	Horizontal spread (wider spread = bigger impact)
Direction of Effect:	Position relative to zero line (right = increases prediction, left = decreases)
Feature Value Relationship:	Color (red/high vs blue/low) — shows how each pushes the prediction
Distribution & Variability:	How the points are spread and clustered
Interactions & Non-linearity:	Complex patterns, asymmetry, or mixed colors on both sides
Consistency of Effect:	Whether points are tightly clustered or widely scattered
Outliers & Rare Effects:	Isolated points far away from the main cluster

1. What each axis means

Y-axis (features):

Features are ordered according to the *mean absolute SHAP value* across all samples.

- Top = most influential variables in the model
- Bottom = least influential

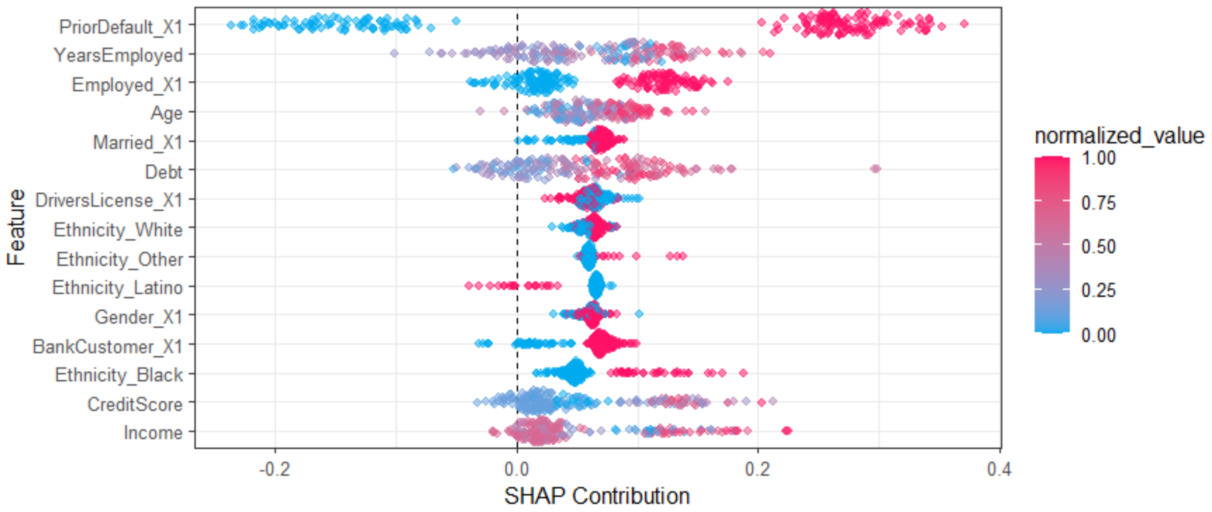
So in our plot, the most important variables are:

  - PriorDefault_X1 (which is the dummy variable for ‘yes’ they defaulted previously)
  - YearsEmployed
  - Employed_X1 (which is the dummy variable for ‘yes’ they are employed)
  - Age
  - Married_X1 (which is the dummy variable for ‘yes’ they are married)

X-axis (SHAP contribution, ϕ_j):

This shows us the *direction* of the effect:

- Values > 0 → push prediction toward class 1
- Values < 0 → push prediction toward class 0



2. Each dot = one observation

Every dot represents one person (row in the dataset). E.g. for Age, we see how it affects prediction for every individual.

3. Colour*

The colour shows the actual feature value:

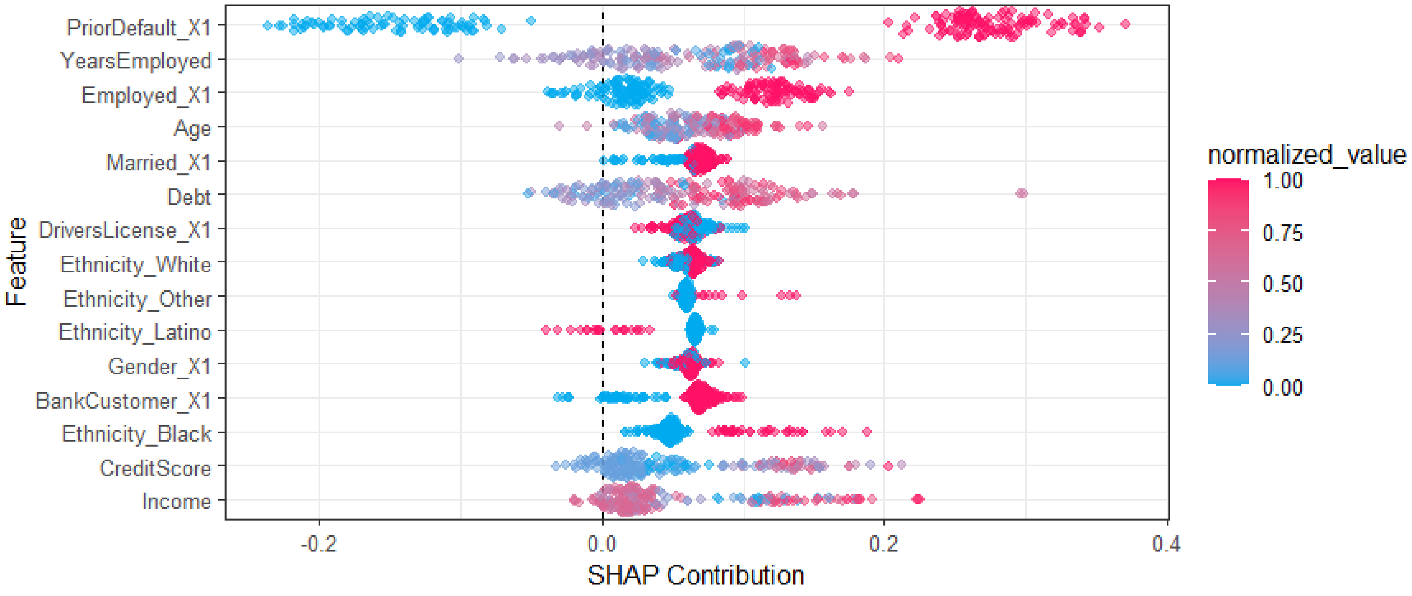
- Blue = low value
- Red = high value

E.g. PriorDefault_X1:

- ✓ Red dots on the right → value = 1 → pushes strongly toward class 1
- ✓ Blue dots on the left → value = 0 → pushes toward class 0

This is exactly what we expect for a binary indicator.

* note: the colour may differ based the package/function used



4. How to interpret a row (example)

E.g. 1: PriorDefault_X1

- Red cluster far right (positive SHAP)
- Blue cluster far left (negative SHAP)

Interpretation: Having a prior default (1) strongly increases likelihood of class 1 (having the credit card application approved). Not having one (0) strongly decreases likelihood of 1. This is the strongest predictor in the model.

E.g. 2: YearsEmployed

- Low values (blue) → slightly negative SHAP
- High values (red) → positive SHAP

Interpretation: More years employed increases probability of class 1. Less experience decreases it.

E.g. 3: Age

- Points are more centred around 0

Interpretation: Age has a weaker overall effect compared to the other top variables.

5. Spread & Distribution

The width of the dot spread tells you about consistency and variability:

- Wide spread: The feature has a large, variable impact on predictions. It sometimes pushes predictions strongly positive, sometimes strongly negative (or both). This usually indicates a very important feature.
- Narrow spread: The feature has a small or consistent impact. If the SHAP values stay close to zero → low importance.
- Clustered tightly: Very little variation in effect across samples.

Strongest impact variables:

- ✓ PriorDefault_X1
- ✓ YearsEmployed
- ✓ Employed_X1

Low impact variables:

- ✓ CreditScore
- ✓ Income